

Senior software engineer and researcher specializing in language modeling, LLM training and evaluation, and tokenization. I have led ML projects end to end and shipped production language models used by hundreds of millions of people.

Work Experience

02/19-Present **Google**

- **Senior Software Engineer** (10/25-Present, Mountain View, CA)
 - Software engineer on Gboard focusing on language modeling and LLM fine-tuning.
 - Research on efficient, on-device language models, LLM pipeline evaluation, and tokenization.
 - Currently working on personalized LLM fine-tuning and prompt optimization as well as federated LLM evaluation.
 - Co-maintainer of SENTENCEPIECE, Google's LLM tokenizer. I helped land a >7x increase in tokenizer throughput through profiling and tokenizer-specific optimizations, which is now used in every Gemini call.
- **PhD Student Researcher** (05/22-9/25, Tokyo, Japan)
 - Researcher focused on CJK language modeling on the Gboard team, part time while I completed my PhD.
 - I led the redevelopment of a training and inference stack for specialized CJK keyboard language modeling. I owned the project from end to end: model training, integration into a larger inference pipeline, keyboard UI/UX improvements, evaluation, etc.
- **Software Engineer** (08/19-04/22, Mountain View, CA)
 - Developed compressed statistical language models for mobile keyboard input.
 - Developed TensorFlow Federated infrastructure and models for privacy-preserving federated analytics.
 - I led a model compression and optimization project that reduced our statistical model's footprint by >70%, which allowed us to move from toy problems to real-world deployment on hundreds of millions of devices.
- **Software Engineering Intern** (02/19-05/19, New York City, NY)
 - Developed a compression algorithm for finite-state transducers used in keyboard language models to reduce space requirements by >90% over the uncompressed version and >58% over the prior production compression scheme (published as a first-author paper).

09/25-10/25 **Chess.com**

- **Machine Learning Consultant** (09/25-10/25, Remote)
 - Modernized and optimized their deep learning stack and developed a state-of-the-art neural chess model that improved their primary internal metric by 84%.
 - I advised on model architecture design, training data curation, and evaluation criteria and brought their team of engineers up to speed on modern deep learning principles.

Education

2022-2025 **PhD Computer Science**, Tokyo Institute of Technology, Tokyo, Japan

- Advisor: Dr. Naoaki Okazaki
- Thesis: *Practical and Theoretical Aspects of Tokenization*
- MEXT Scholar (文部科学省奨学金)

2016-2018 **MS Computer Science**, Yonsei University, Seoul, South Korea

- Advisor: Dr. Yo-Sub Han
- Thesis: *Efficient Algorithms for Two Parsing Problems on Probabilistic Finite Automata*

2011-2015 **BS Discrete Mathematics**, Georgia Institute of Technology, Atlanta, GA

- Minor in Korean
- Advisor: Dr. Anton Leykin
- Thesis: *Straight Line Programs and Automatic Differentiation in Python*

Skills

- **Programming Languages:** Python, C++, Julia
- **Tools:** PyTorch, Flux.jl, OpenFst, TensorFlow Federated
- **Human Languages:** English, Korean

Publications (*denotes primary authorship)

1. **Tokenization as Finite-State Transduction**
 - Marco Cognition*, Naoaki Okazaki. Computational Linguistics.
2. **The bread emoji Team's Submission to the 2025 FedCSIS Predicting Chess Puzzle Difficulty Challenge**
 - Tyler Woodruff, Luke Imbing, Marco Cognition. FedCSIS 2025.
 - Our submission placed 2nd (out of 73 teams) and won a \$500 prize.
3. **Pitfalls, Subtleties, and Techniques in Automata-Based Subword-Level Constrained Generation**
 - Marco Cognition*, David Pohl*, Junyoung Lee, Naoaki Okazaki. TokShop 2025.
4. **Jamo-Level Subword Tokenization in Low-Resource Korean Machine Translation**
 - Junyoung Lee*, Marco Cognition*, Sangwhan Moon, Naoaki Okazaki. LoResMT 2025.
5. **Distributional Properties of Subword Regularization**
 - Marco Cognition*, Vilém Zouhar, Naoaki Okazaki. EMNLP 2024.
6. **The bread emoji Team's Submission to the IEEE BigData 2024 Cup: Predicting Chess Puzzle Difficulty Challenge**
 - Tyler Woodruff, Oleg Filatov, Marco Cognition. IEEE Big Data 2024.
 - Our submission won 1st place (out of 143 teams) and a \$1000 prize.

7. **Two Counterexamples to *Tokenization and the Noiseless Channel***
- Marco Cognition*, Vilém Zouhar, Sangwhan Moon, Naoaki Okazaki. LREC-COLING 2024.
8. **Parameter-Efficient Korean Character-Level Language Modeling**
- Marco Cognition*, Sangwhan Moon, Lawrence Wolf-Sonkin, Naoaki Okazaki. EACL 2023.
9. **SoftRegex: Generating Regex from Natural Language Descriptions using Softened Regex Equivalence**
- Jun-U Park, Sang-Ki Ko, Marco Cognition, Yo-Sub Han. EMNLP 2019.
10. **On the Compression of Lexicon Transducers**
- Marco Cognition*, Cyril Allauzen, Michael Riley. FSMNLP 2019.
11. **Online Infix Probability Computation for Probabilistic Finite Automata**
- Marco Cognition*, Yo-Sub Han, Soon Chan Kwon. ACL 2019.
12. **Incremental Computation of Infix Probabilities for Probabilistic Finite Automata**
- Marco Cognition*, Yo-Sub Han, Soon Chan Kwon. EMNLP 2018.
13. **Online Stochastic Pattern Matching**
- Marco Cognition*, Yo-Sub Han. CIAA 2018.

Preprints and Non-Archival Publications (*denotes primary authorship)

1. **Decoding-Free Sampling Strategies for LLM Marginalization**
- David Pohl*, Marco Cognition*, Junyoung Lee, Naoaki Okazaki.
- <https://arxiv.org/abs/2510.20208>
2. **Tutorial: φ -Transductions in OpenFst via the Gallic Semiring**
- Marco Cognition*, Cyril Allauzen.
- <https://arxiv.org/abs/2506.17942>
3. **An Analysis of BPE Vocabulary Trimming in Neural Machine Translation**
- Marco Cognition*, Tatsuya Hiraoka, Naoaki Okazaki, Rico Sennrich, Yuval Pinter.
- <https://arxiv.org/abs/2404.00397>
4. **A Call for an Open Tokenizer Benchmark** (extended abstract, non-archival)
- Marco Cognition*. TokShop 2026.

Conference Talks

1. **LotteryTickets.jl: Sparsify Your Flux Models**
- JuliaCon 2023 (Boston, USA)
- <https://www.youtube.com/watch?v=ZmcaUyZLi4Q>
- <https://github.com/mcognition/LotteryTickets.jl>

Invited Talks

1. **Subword Tokenization Meets Formal Language Theory**
- Invited Tutorial at Developments in Language Theory (DLT) (August 2025, Seoul)
- https://github.com/mcognition/subword_tokenization_meets_formal_language_theory
2. **The Tokenization Landscape**
- National Institute of Advanced Industrial Science and Technology Artificial Intelligence Research Center's Knowledge and Information Research Team (AIST AIRC-KIRT) (March 2024, Tokyo)

Service

- **The Token ##ization Discord** - Founder and Organizer²⁰²⁵⁻
- **Reviewer** - *ACL^{2024, 2025}, TCS²⁰²⁵, AAAI²⁰²⁵, TokShop²⁰²⁵, NLP-OSS²⁰²³, JuliaCon^{2022, 2023}
- **Maclay High School** - STEM Council External Advisor²⁰²⁵⁻²⁰²⁶
- **Women in Science Japan's Machine Learning Summer School for Scientists** - Mentor²⁰²⁵
- **Seminars on Formal Languages and Neural Networks (FLaNN)** - Co-organizer²⁰²²⁻²⁰²⁵
- **The Gradient** (<https://thegradients.pub/>) - Editorial Board²⁰²¹⁻²⁰²⁴
- **FSU ACM Programming Contest** - Question Writer^{2020, 2021 (x2), 2022}
- **Hackbright Academy** - Volunteer Mentor^{2020 (x2), 2021}
- **ACM International Collegiate Programming Contest (Korea Regional)** - Question Writer^{2017, 2018}

Selected Side-Projects

1. **blunder.clinic**[🏆] - *Realistic Daily Chess Puzzles*
- Chess puzzle app driven by human-like models that identifies positions in which players at a specific skill level made a mistake even though the best move should have been within their ability to find.
- Implemented an end-to-end neural chess model training and inference pipeline that matches the current state of the art.
- Several thousand weekly active users.
- Featured[🏆] on the front page of Lichess.